A Dynamic Structural Model of Subscription Churn

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Roadmap

1. Setting
2. Data
3. Model
4. Next steps

Setting

Each month, a user chooses whether to **continue subscribing** to a music streaming platform.

Key Assumption: Platform Capital

The more accumulated hours a user has on the platform, the more *platform capital* they have:

- E.g., playlists and algorithmic learning of music tastes
- More time spent $\Rightarrow$ higher utility from remaining subscribed

Source: KKBox (Kaggle) | **Period:** Jan 2015 – Feb 2017

User-month panel sample

- 753 users, mean length of 9.1 months
- **Eligibility**: registered within the observation window; at least one transaction
- **Variables**: month hours played, auto-renewal status, subscription price, user demographics

Notable Features

- Auto-renewal and price are *constant* within a user's spell
- Price varies *across* users (e.g., 99, 149, 180 TWD)
- Paused users accumulate zero hours; churn is absorbing

Model Overview

- Each 30-day period t after joining the platform, user i observes their **state** Ω_{it} and **shock** $\{\varepsilon_{ita}\}_{a=0}^{2} \sim$ T1EV
- User chooses an action $a \in \{0, 1, 2\}$ corresponding to **Churn**, **Active**, **Pause**
- State $\Omega_{it} = (R_i, p_i, h_{it}, a_{i,t-1})$ is a collection of:
  - Auto-renew indicator R_i (*constant*)
  - Monthly subscription price p_i (*constant*)
  - Accumulated hours h_{it} (*endogenous*)
  - Inertia term $\mathbb{1}\{a_{it} = a_{i,t-1}\}$ (*endogenous*)

Current-Period Utilities

Actions: $a \in \{0, 1, 2\} \longleftrightarrow$ {Churn, Active, Pause}

Current-Period Utilities

$$u_0(\Omega_{it}) = 0 + \varepsilon_{it0} \qquad \text{(Churn)}$$

$$u_1(\Omega_{it}) = \alpha R_{it} - \beta p_{it} + \eta \mathbb{1}\{a_{i,t-1} = 1\} + \gamma h_{it} + \varepsilon_{it1} \qquad \text{(Active)}$$

$$u_2(\Omega_{it}) = \bar{u}_2 + \eta \mathbb{1}\{a_{i,t-1} = 2\} + \varepsilon_{it2} \qquad \text{(Pause)}$$

Assume a latent-class structure on all parameters $(\alpha, \beta, \eta, \gamma)$ to address **unobserved heterogeneity** (e.g., *music lovers* vs. *casual users*). Omitted for now.

Laws of Motion

Exogenous states (R_i, p_i): Constant over months per user.

Inertia: Deterministic transition based on last month's action.

Accumulated hours h_{it}: Continuous values *discretized* into H bins (can think as "tiers" of experience)

- E.g., 0–10, 10–50, 50–100, 100–300, 300+ hrs (use empirical quantiles?)
- Assume monthly hours are stochastic from the user's perspective. Transition probabilities between discrete states estimated from **empirical distribution of transitions among active users**.

Action-specific transition probabilities:

- Active Users: Hours accumulate and experiences inertia.
- Paused Users: **Do not accumulate hours** (probability of transitioning between hour bins is zero). Can still transition between inertia states.
- Churned Users: Enter an **absorbing terminal state** with **0 utility** for all future periods.

Bellman Equation

Choice-Specific Value Functions

The deterministic component of the expected discounted utility of choosing action a and behaving optimally thereafter are

$$\begin{aligned}
\bar{V}_0(\Omega_{it}) &= 0 \\
\bar{V}_1(\Omega_{it}) &= u_1(\Omega_{it}) + \delta \mathbb{E}[V(\Omega_{i,t+1}) \mid \Omega_{it}, a = 1] \\
\bar{V}_2(\Omega_{it}) &= u_2(\Omega_{it}) + \delta \mathbb{E}[V(\Omega_{i,t+1}) \mid \Omega_{it}, a = 2]
\end{aligned}$$

Ex-Ante Value Function

Integrating over the Type-I Extreme Value shocks $\{\varepsilon_{ita}\}$ yields

$$V(\Omega_{it}) = \gamma_{EM} + \log\left(1 + \exp(\bar{V}_1(\Omega_{it})) + \exp(\bar{V}_2(\Omega_{it}))\right)$$

$$\Pr(a \mid \Omega_{it}) = \frac{\exp(\bar{V}_a(\Omega_{it}))}{1 + \exp(\bar{V}_1(\Omega_{it})) + \exp(\bar{V}_2(\Omega_{it}))}$$

Summary & Next Steps

Model Features

- Dynamic discrete choice
- Endogenous platform capital via accumulated hours
- Inertia / switching costs
- Latent class heterogeneity

Next Steps

- Estimation with nested fixed point
- Counterfactuals (e.g., price changes, removing auto-renew)
- Compare with non-structural probability mixture model (BG/BB)